

FAQs for the Mini-Project

What is the Grading Scheme for the Mini-Project?

30% for coming up with an interesting problem based on your dataset

30% for data extraction, and cleaning for exploratory data analysis/visualization to gather relevant insights

30% for the use of machine learning techniques to solve a specific problem

10% for learning and implementing something new and trying something beyond the course

If you are attempting something different, especially something that you think do not fit naturally into the grading scheme, feel free to discuss with us.

What is an "interesting problem" based on the dataset?

Talk to us if you are not sure. We will help you choose something interesting, in case you haven't been able to decide. It should not be something that you can solve by copy-pasting the LinearRegression or DecisionTreeClassifier codes from the regular course material. There should be something beyond that, for which you will have to learn something new, or apply some new technique. If you are unsure whether your problem is interesting, email us.

Warning : In quest for "interesting problem", please do not attempt something that you can't finish in time.

What do you mean by "data extraction, curation and cleaning?"

The data sources given to you may not have the data you want in the exact format you want it. Some of the sources only offer you with an API, while some others have multiple data tables. Extracting, curating and/or joining the data means understanding your data sources, learning about the APIs if needed, and procuring the data in a way that's most suited to the problem you want to solve. This is an important part of any DS/ML project.

The dataset you extract from the sources may not be in the proper format to solve the problem you targeted. Preparing means cleaning the data, resizing/reshaping the data, removing outliers (if necessary), balancing imbalanced classes (if necessary), grouping the rows/columns as necessary, etc. This is also an important part of any DS/ML project.

How much of Visualization should be presented?

Do not spend the bulk of time on cool visualization tools, spend more time on the analysis and machine learning. Do standard exploration of the data, and standard statistical visualizations, as done during the course, just to understand your data well enough, and to gather relevant insights that you can present in connection with your problem. You DO NOT need to produce data dashboards and cool web interfaces to do an impressive project.

Warning : In quest for "cool presentation", please do not try something that takes too much time to learn.

How much of machine learning techniques should I use for the project?

This is important, but not the ONLY focus of your project. You may use any tool and technique that you have seen during the course, for Regression and Classification, but make sure that it is actually required to solve the problem you formulate on the dataset you choose. If you want something simple, stick to Scikit-Learn as your ML toolbox. You may also choose to use new models that you have not seen in the course, like Random Forest for regression, Naive Bayes for classification, or the Xtra Module techniques like clustering, anomaly detection, etc. This will also satisfy the criteria of "learning something new" in parallel.

Warning : In quest for "quick impression", please do not try complex tools that takes too much time to learn.

What do you mean by "learning something new" beyond the course?

The goal for the mini-project is to make you learn something new. Try to use a new ML model for regression or classification, beyond what we have already covered in the course, or apply a model for clustering or anomaly detection. That's the quickest way to prove you learned something new. You may also want to explore a new visualization tool (like Plotly), or a new technique for data preparation (like resampling), or explore additional/extra data beyond what have been suggested for your project. Basically, anything that we haven't already taught you!

Warning : In quest for "quick impression", please do not try to learn too many new things at the same time.

What is the format for the final Presentation?

You will present a 5-minute presentation (excluding Q&A). You may format it as a PPT/PDF/Google Slides presentation on the problem you chose, along with some snippets of your code from Jupyter Notebook/Google Colab, and relevant visualizations. You may also feel free to "demonstrate" any tool or app that you built based on the dataset or your problem. You DO NOT need to show raw code during the presentation, but you may of course, if you want. The idea for the presentation will be a complete presentation, explaining all the steps from your motivation to your data-driven solution to the problem. Pay attention to the grading scheme to judge what to present.

Warning : In quest for "quick impression", please do not try to "produce a film" with cool animations and stuff. That's not the point! ;-)

Who will grade my Mini-Project?

The Presentations will be graded finally by Prof. Leonard, Prof Kedar and TAs: Andre and Amrita.

What will I finally submit for the Mini-Project?

You will have to submit the PPT/PDF slides used for the Presentation, all your codes (Jupyter Notebooks, Python codes, Visualization codes, etc.), with reference to the resources you used during the project. If you build an application, a visualization tool, a dashboard or a website, you may also submit the link to that.

Note : Guidelines on Presentation and Submission of Project will be communicated closer to the date.